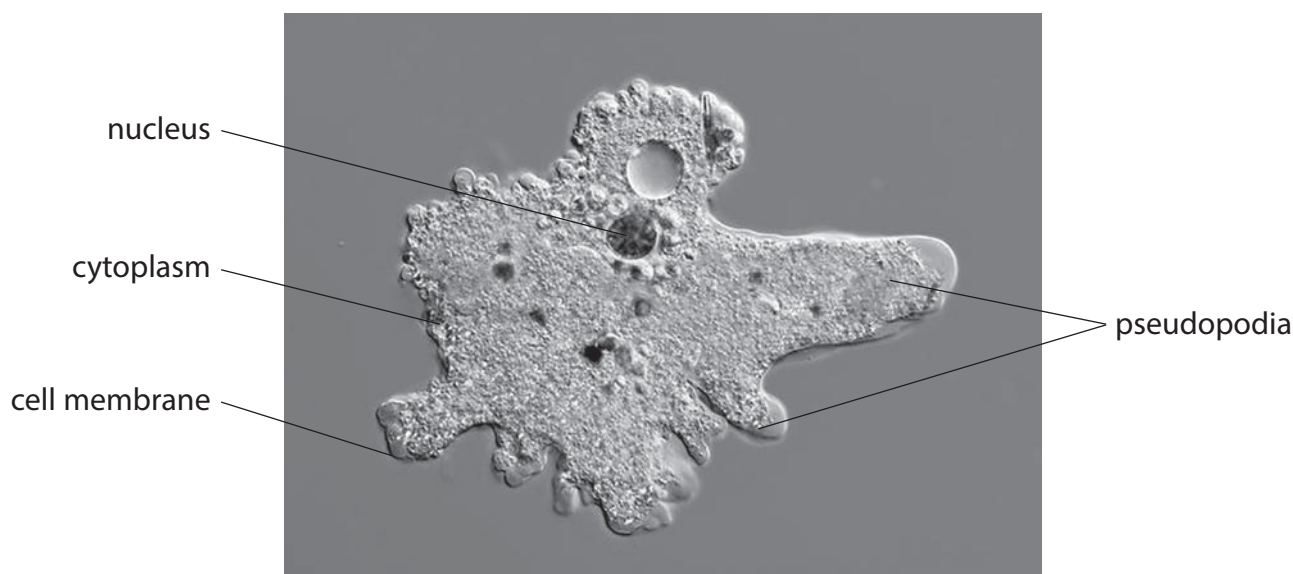


5 An amoeba is a single-celled organism that lives in water.

The photograph shows an amoeba, as seen using a light microscope.



© GERD GUENTHER / SCIENCE PHOTO LIBRARY

Substances in the water can enter the amoeba by a variety of transport mechanisms.

(a) The table shows some features of transport mechanisms.

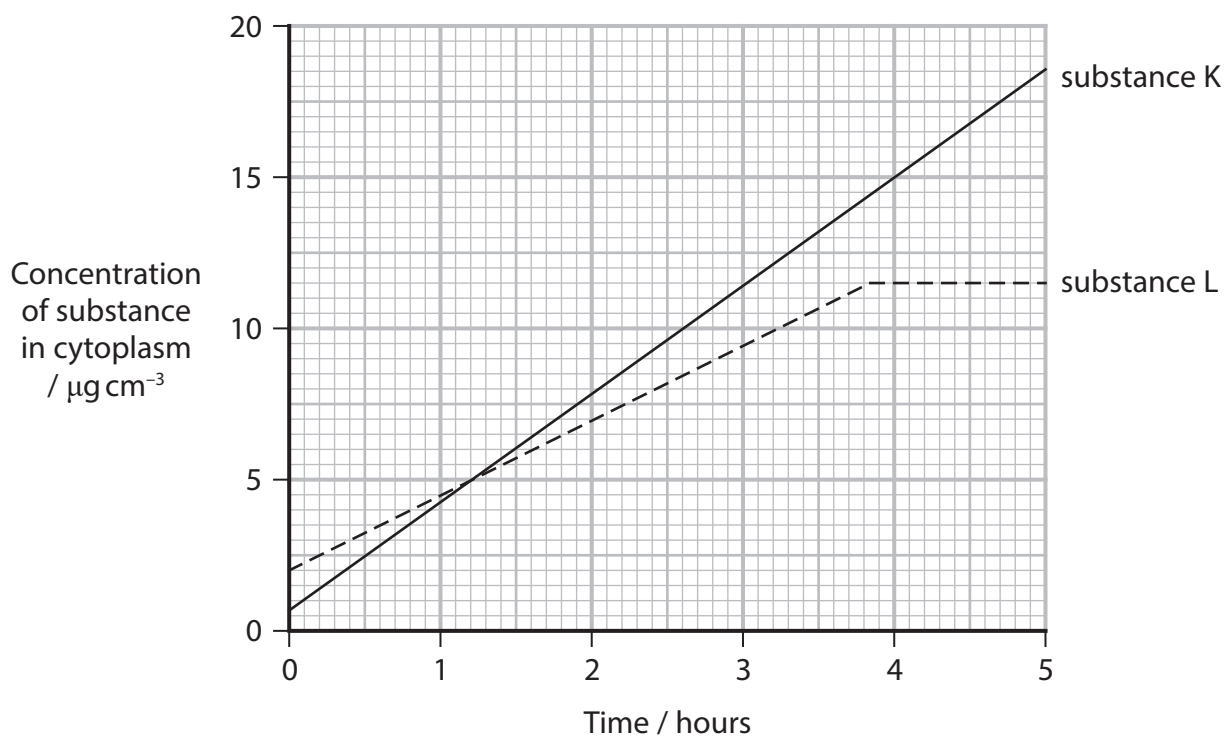
Which feature is true for active transport only, facilitated diffusion only, both active transport and facilitated diffusion, or not true for both active transport and facilitated diffusion?

(3)

Feature	Transport mechanism			
	active transport only	facilitated diffusion only	both active transport and facilitated diffusion	not true for both active transport and facilitated diffusion
passive process	☒	☒	☒	☒
membrane proteins involved	☒	☒	☒	☒
direction of transport can be up the concentration gradient	☒	☒	☒	☒



(b) The graph shows the uptake of two substances, substance K and substance L, into the cytoplasm of an amoeba in water at a temperature of 18 °C.



(i) Explain the differences in the uptake of substance K and substance L.

(2)

.....

.....

.....

.....

.....

.....

(ii) On the graph, draw a line to show the uptake of substance L at 10 °C.

(2)

(c) Pseudopodia are formed by cytoplasm flowing towards the membrane and changing its shape.

- (i) Explain why the membrane is able to change its shape when cytoplasm flows towards it.

(2)

- (ii) Explain how the uptake of substances would be affected if the amoeba increased its number of pseudopodia.

(2)

(Total for Question 5 = 11 marks)

